

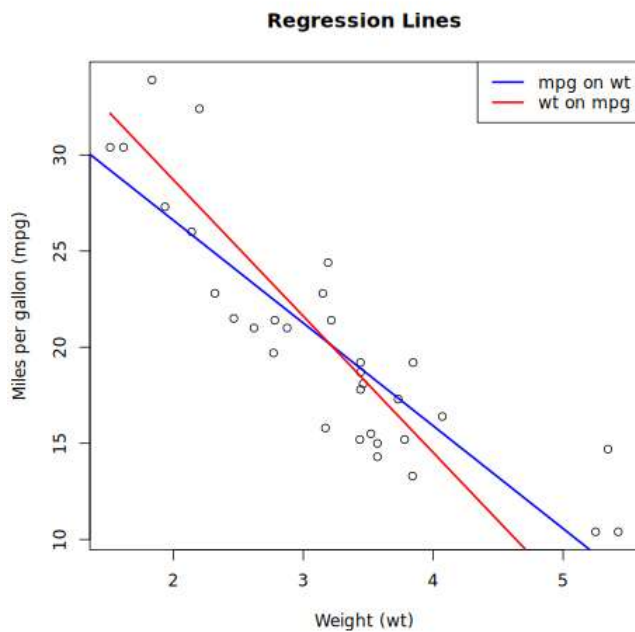
For Number 4 (Letter C)

c.)

Code:

```
1 data(mtcars)
2
3 # Fit models
4 model1 <- lm(mpg ~ wt, data = mtcars)
5 model2 <- lm(wt ~ mpg, data = mtcars)
6
7 # Plot
8 plot(mtcars$wt, mtcars$mpg,
9      xlab = "Weight (wt)",
10     ylab = "Miles per gallon (mpg)",
11     main = "Regression Lines")
12
13 # Line 1: mpg on wt
14 abline(model1, col = "blue", lwd = 2)
15
16 # Line 2: wt on mpg (rearranged to mpg vs wt form)
17 # Solve: wt = 6.047 - 0.141 mpg → mpg = (6.047 - wt)/0.141
18 curve((6.047 - x)/0.141, add = TRUE, col = "red", lwd = 2)
19
20 legend("topright",
21       legend = c("mpg on wt", "wt on mpg"),
22       col = c("blue", "red"),
23       lwd = 2)
```

Output:



For Number 5:

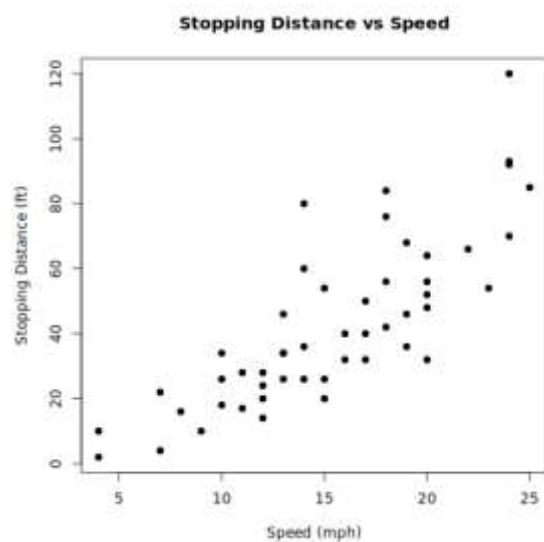
a.)

Code:



```
1 # Load dataset
2 data(cars)
3
4 # Scatter plot
5 plot(cars$speed, cars$dist,
6       main = "Stopping Distance vs Speed",
7       xlab = "Speed (mph)",
8       ylab = "Stopping Distance (ft)",
9       pch = 19)
```

Output:



b.)

Code:

```
1 # Load data
2 data(cars)
3 #letter b
4 b = "letter B"
5 print(b)
6 quad_model <- lm(dist ~ speed + I(speed^2), data = cars)
7 summary(quad_model)
_
```

Output:

Output

```
[1] "letter B"

Call:
lm(formula = dist ~ speed + I(speed^2), data = cars)

Residuals:
    Min       1Q   Median       3Q      Max
-28.720  -9.184  -3.188   4.628  45.152

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.47014    14.81716   0.167   0.888
speed         0.91329     2.03422   0.449   0.656
I(speed^2)    0.09996     0.06597   1.515   0.136

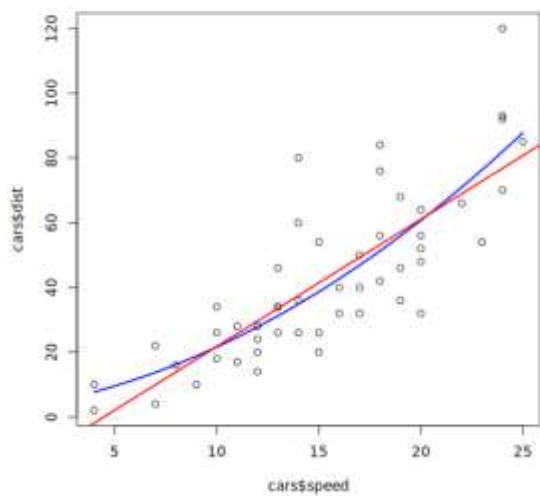
Residual standard error: 15.18 on 47 degrees of freedom
Multiple R-squared:  0.6673,    Adjusted R-squared:  0.6532
F-statistic: 47.14 on 2 and 47 DF,  p-value: 5.852e-12
```

c.)

Code:

```
9 #letter c
10 c = "letter C"
11 print(c)
12
13 # Plot
14 plot(cars$speed, cars$dist)
15
16 # Fit model
17 quad_model <- lm(dist ~ speed + I(speed^2), data = cars)
18
19 # Add curve
20 speed_seq <- seq(min(cars$speed), max(cars$speed), length.out = 100)
21 pred_quad <- predict(quad_model, newdata = data.frame(speed = speed_seq))
22
23 lines(speed_seq, pred_quad, col = "blue", lwd = 2)
24
```

Output:



d.)

Code:

```
26 #letter d
27 d = "letter D"
28 print(d)
29 predict(quad_model, newdata = data.frame(speed = c(45, 80)))
```

Output:

```
[1] "letter D"
      1      2
245.9857 715.2727
```

e.)

Code:

```
31 #letter e
32 e = "letter E"
33 print(e)
34 lin_model <- lm(dist ~ speed, data = cars)
35
36 # Add linear line
37 abline(lin_model, col = "red", lwd = 2)
38
39 summary(lin_model)
```

Output:

```
[1] "letter E"

Call:
lm(formula = dist ~ speed, data = cars)

Residuals:
    Min       1Q   Median       3Q      Max
-29.069  -9.525  -2.272   9.215  43.201

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791     6.7584  -2.601  0.0123 *
speed         3.9324     0.4155   9.464 1.49e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared:  0.6511,    Adjusted R-squared:  0.6438
F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12

[Execution complete with exit code 0]
```